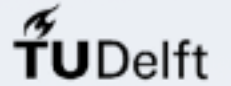




Probability and Statistics: Lecture 11

Hypothesis Testing

PRIME



Programme

- Hypothesis testing
- Test statistic and p -value
- Asymptotic test for a proportion
- t -test for the mean



Sections 25, 26.1, 26.2, 27.1, and 27.2

Hypothesis Testing



Models

Hypothesis Testing



At a party you want to test whether tapping a can of soda makes the drink less likely to 'foam over'. Let p_{tap} and $p_{\text{no tap}}$ be the respective probabilities of foaming over for tapping and not tapping a can. What is your null hypothesis and what is your alternative hypothesis?

- A. $H_0 : p_{\text{tap}} = p_{\text{no tap}}, H_1 : p_{\text{tap}} < p_{\text{no tap}}$
- B. $H_0 : p_{\text{tap}} < p_{\text{no tap}}, H_1 : p_{\text{tap}} = p_{\text{no tap}}$
- C. $H_0 : p_{\text{tap}} > p_{\text{no tap}}, H_1 : p_{\text{tap}} = p_{\text{no tap}}$
- D. $H_0 : p_{\text{tap}} = p_{\text{no tap}}, H_1 : p_{\text{tap}} > p_{\text{no tap}}$



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Introduction: what is a statistical test?



You have two hypotheses H_0 and H_1 and you would like to test whether the unknown data generating process satisfies H_0 or H_1 . The goal is to find enough evidence to be sure to reject H_0 .

Can you think of an example of H_0 and H_1 ?



The three steps of hypothesis testing



1. Formulate H_0 and H_1
2. Do the experiment
3. Calculate whether results justify rejecting H_0

Results:

Do not reject H_0

Reject H_0

Insufficient evidence to support H_1

H_1 true “beyond reasonable doubt”

Reasoning: checking defective proportion



A manufacturer of integrated circuits claims that 1% of its integrated circuits are defective. Based on previous experiences with this manufacturer you think the defective proportion is higher.

Therefore, you want to test the manufacturer's claim.

Let θ be the unknown 'true' defective proportion.

What would you choose as H_0 and H_1 ?

The appropriate hypotheses are:

$$H_0 : \theta = 0.01$$

$$H_1 : \theta > 0.01$$

Type I and Type II errors



Probability



Models

Type I and Type II errors



Our decision based on the test



Do not reject H_0

Reject H_0

H_0 is true

Correct

Type I error

H_1 is true

Type II error

Correct

True state of nature





Introduction: Type I and Type II errors

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You went to a party and drank some beers. On your way home, the police stop you for an alcohol check. According to the breath analyzer you are sober and the police let you go. What kind of error are the police making?

- A. Type I
- B. Type II
- C. The police didn't make an error
- D. I can not tell from the information I have so far





Introduction: Type I and Type II errors

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Significance Level

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Definition:

The significance level α is the largest acceptable probability of committing a type I error.

We speak of “performing the test at level α ”, as well as “rejecting H_0 in favor of H_1 at level α ”.



Significance Level

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	Do not reject H_0	Reject H_0
H_0 is true	Correct	Type I error
H_1 is true	Type II error	Correct

Significance level of the test:

$$\alpha = P_{H_0}(\text{reject}, H_0) = P_{H_0}(\text{type I error})$$

We would like α to be small, i.e., close to 0.

Example: $\alpha = 0.05$.

t -test for the mean



Estimation



Probability



Models



Reasoning: error and p-value

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Consider the following two statements:

1. A Type II error occurs if H_0 is falsely rejected.
2. The p -value is the probability that H_0 is true.

Which statement is/are true?

- A. None
- B. Only 1
- C. Only 2
- D. Both





Reasoning: error and p-value

14

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1. A Type II error occurs if H_0 is falsely rejected.
2. The p -value is the probability that H_0 is true.

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Introduction: hypothesis testing for a proportion

15

Last year, 15% of voters voted for the P&S party. This year, there will be another election. You wonder if this proportion has changed. You organize a poll, asking $n = 1000$ people whether they intend to vote P&S (yes/no).

We want to test $H_0 : p = 0.15$ against $H_1 : p \neq 0.15$.

Suppose out of 1000 people, 177 say they intend to vote P&S.

Call X the number of people who answered yes.

Question 1: Assuming H_0 , what is the distribution of X ?





Introduction: hypothesis testing for a proportion

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See you next lecture

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